Drive systems & mechanisms

ENGINEERING WITH MR KELLY

National 5 · Mechanisms & structures · Worksheet

Name _____ Class _____ Date _____

By the end of this worksheet I can…

- ☐ name the four types of motion and give examples
- ☐ calculate velocity ratio, output speed and torque in gear trains
- ☐ describe the effect of an idler gear
- ☐ design a compound gear train to give a required speed
- ☐ describe ways to reduce friction and wear

I DO — Worked example

Read how a full-mark answer is built

An idler gear B is placed between gears A and C.

1 State the effect on the output speed if the number of teeth on idler B is reduced. (1 mark)

Full-mark answer each ✓ is one mark

There is no effect on the output speed; an idler only changes the direction of rotation ✓.

Why marks are not awarded

"It reverses the direction of gear C."

0 marks: the question asks about speed.

WE DO — Guided practice

Use the starters and word bank to help you

Moving parts in a drive system wear quickly.

2 Describe one way to reduce wear. (1 mark)

Sentence starters

One … is …, which means …

Another … is …, so …

Word bank

harder · grease · materials · reduce · self-lubricating · friction · bearings · lubricate

Tip: 1 marks = 1 separate, developed points.

YOU DO **On your own** No notes. Check your answers afterwards.

3 Quick check: tick one box for each.

a) A pneumatic piston moving in and out shows...
☐ Rotary motion ☐ Oscillating motion ☐ Reciprocating motion ☐ Linear only

b) A 20-tooth driver turns a 60-tooth gear. The VR is...
☐ 1:3 ☐ 3:1 ☐ 20:1 ☐ 60:1

c) A 1500 rev/min, 52-tooth gear drives a 12-tooth gear. Output speed is about...
☐ 346 rev/min ☐ 6500 rev/min ☐ 1500 rev/min ☐ 780 rev/min

d) Reducing the teeth on an idler gear makes the output...
☐ Faster ☐ Slower ☐ No different ☐ Reverse

4 Key terms: write the letter of the matching description next to each term.

1. **Reciprocating motion**	____	**A** Backwards and forwards in a straight line.
2. **Oscillating motion**	____	**B** Two gears on one shaft; total VR = VR1 × VR2.
3. **Velocity ratio**	____	**C** Speed of input ÷ speed of output = driven teeth ÷ driver teeth.
4. **Gear speed method**	____	**D** Swinging backwards and forwards in an arc.
5. **Idler gear**	____	**E** Input speed × input teeth = output speed × output teeth.
6. **Compound gear train**	____	**F** Makes driver and driven turn the same way; doesn't change the VR.

A motorised winch is driven by a motor at 1200 revs min^{-1}. The drum must turn at 25 revs min^{-1}.

5 Calculate the velocity ratio required. (2 marks)

A 20-tooth driver gear meshes with a 60-tooth driven gear. The motor turns at 1500 revs min⁻¹.

6 Calculate the velocity ratio and the output speed. (3 marks)

A compound gear train: A (15 teeth) drives B (45 teeth); C (10 teeth, on the same shaft as B) drives D (50 teeth). Input 3000 revs min⁻¹.

7 Calculate the output speed. (3 marks)

Check and reflect

Mark your answers with the answer sheet. Go back to the checklist on the site and re-rate each “I can” statement above. Which one still needs work?